

Python

100 questions · compiled from 2 files

SCRIPTING

1. What is the main purpose of the `if __name__ == '__main__':` block in Python?

1. To define a class
2. To ensure code runs only when the file is executed directly
3. To import external modules
4. To create a lambda function

Answer 2 · To ensure code runs only when the file is executed directly

SCRIPTING

2. Which module is commonly used to access command-line arguments in Python?

1. os
2. sys
3. math
4. json

Answer 2 · sys

SCRIPTING

3. What does `sys.argv[0]` usually contain?

1. The first user argument
2. The Python version
3. The script name or path
4. The current working directory

Answer 3 · The script name or path

SCRIPTING

4. Which function is used to read input from a user in Python 3?

1. scan()
2. read()
3. input()
4. get()

Answer 3 · input()

SCRIPTING

5. What is the type of the value returned by `input()`?

1. int
2. float
3. str
4. bool

Answer 3 · str

SCRIPTING

6. Which module is commonly used to work with CSV files in Python?

1. json
2. csv
3. pickle
4. sys

Answer 2 · csv

SCRIPTING

7. What is the safest way to open a file so it closes automatically after use?

1. open() only
2. try without except
3. with open(...) as f
4. close() before reading

Answer 3 · with open(...) as f

SCRIPTING

8. Which file mode is used to append to an existing file?

1. r
2. w
3. a
4. x

Answer 3 · a

SCRIPTING

9. What happens when you open a file in w mode that already exists?

1. An error is raised
2. The file is deleted permanently
3. The file is overwritten
4. The file becomes read-only

Answer 3 · The file is overwritten

SCRIPTING

10. Which statement correctly imports only the sqrt function from math?

1. import math.sqrt
2. from math import sqrt
3. include sqrt from math
4. using math.sqrt

Answer 2 · from math import sqrt

NUMPY

11. What is the main advantage of NumPy arrays over Python lists for numerical work?

1. They can store only strings
2. They are faster and more memory-efficient for numeric operations
3. They automatically sort values
4. They do not support indexing

Answer 2 · They are faster and more memory-efficient for numeric operations

NUMPY

12. Which function creates a NumPy array from a Python list?

1. `np.list()`
2. `np.array()`
3. `np.make()`
4. `np.convert()`

Answer 2 · `np.array()`

NUMPY

13. What is the shape of a NumPy array created by `np.array([[1,2],[3,4],[5,6]])`?

1. (2, 3)
2. (3, 2)
3. (3,)
4. (2,)

Answer 2 · (3, 2)

NUMPY

14. Which NumPy function creates an array of zeros?

1. `np.empty()`
2. `np.ones()`
3. `np.zeros()`
4. `np.null()`

Answer 3 · `np.zeros()`

NUMPY

15. What is the result of `np.array([1,2,3]) + 2`?

1. [1,2,3,2]
2. [3,4,5]
3. Error
4. [2,4,6]

Answer 2 · [3,4,5]

NUMPY

16. Which attribute gives the number of dimensions of a NumPy array?

1. `size`
2. `shape`
3. `ndim`
4. `dtype`

Answer 3 · `ndim`

NUMPY

17. What does broadcasting allow in NumPy?

1. Sorting arrays automatically
2. Performing operations on arrays of compatible shapes
3. Converting arrays to lists
4. Removing duplicate values

Answer 2 · Performing operations on arrays of compatible shapes

NUMPY

18. Which function generates evenly spaced values within a given interval?

1. `np.range()`
2. `np.linspace()`
3. `np.slice()`
4. `np.split()`

Answer 2 · `np.linspace()`

NUMPY

19. What is the output type of `np.mean(np.array([1,2,3,4]))`?

1. list
2. tuple
3. NumPy scalar
4. dict

Answer 3 · NumPy scalar

NUMPY

20. Which operation performs element-wise multiplication on two NumPy arrays of the same shape?

1. `a + b`
2. `a * b`
3. `a.dot(b)`
4. `a @ b` always returns element-wise

Answer 2 · `a * b`

PANDAS

21. Which Pandas data structure is two-dimensional and tabular?

1. Series
2. DataFrame
3. ArrayFrame
4. PanelList

Answer 2 · DataFrame

PANDAS

22. Which function is commonly used to read a CSV file into Pandas?

1. `pd.load_csv()`
2. `pd.read_csv()`
3. `pd.csv_read()`
4. `pd.open_csv()`

Answer 2 · `pd.read_csv()`

PANDAS

23. What does `df.head(3)` return?

1. The last 3 rows
2. A summary of 3 columns
3. The first 3 rows
4. The shape of the DataFrame

Answer 3 · The first 3 rows

PANDAS

24. Which attribute gives the column labels of a DataFrame?

1. df.rows
2. df.columns
3. df.indexes
4. df.labels()

Answer 2 · df.columns

PANDAS

25. What is the difference between loc and iloc in Pandas?

1. loc uses integer positions, iloc uses labels
2. Both do the same thing always
3. loc uses labels, iloc uses integer positions
4. iloc only works on Series

Answer 3 · loc uses labels, iloc uses integer positions

PANDAS

26. Which method is used to display summary statistics for numeric columns?

1. df.info()
2. df.describe()
3. df.stats()
4. df.summary()

Answer 2 · df.describe()

PANDAS

27. What does df['col'].isnull().sum() typically compute?

1. The number of unique values in col
2. The number of missing values in col
3. The total of all numeric values in col
4. The number of rows in the DataFrame

Answer 2 · The number of missing values in col

PANDAS

28. Which method removes rows with missing values by default?

1. dropna()
2. remove_na()
3. fillna()
4. clearna()

Answer 1 · dropna()

PANDAS

29. What does df.groupby('A')['B'].sum() do?

1. Sorts column B
2. Groups rows by A and sums values in B for each group
3. Adds columns A and B
4. Filters rows where A equals B

Answer 2 · Groups rows by A and sums values in B for each group

PANDAS

30. Which method can be used to sort a DataFrame by a column?

1. order_by()
2. sort()
3. sort_values()
4. arrange()

Answer 3 · `sort_values()`

ADVANCED TOPICS

31. What is a list comprehension primarily used for?

1. Defining classes
2. Creating lists concisely from iterables
3. Importing modules
4. Handling exceptions

Answer 2 · `Creating lists concisely from iterables`

ADVANCED TOPICS

32. Which list comprehension creates squares of even numbers from 0 to 4?

1. `[x**2 for x in range(5)]`
2. `[x for x in range(5) if x**2 % 2 == 0]`
3. `[x**2 for x in range(5) if x % 2 == 0]`
4. `[x%2 for x in range(5)]`

Answer 3 · `[x**2 for x in range(5) if x % 2 == 0]`

ADVANCED TOPICS

33. What is a lambda function in Python?

1. A named loop
2. An anonymous small function
3. A file handler
4. A debugging tool

Answer 2 · `An anonymous small function`

ADVANCED TOPICS

34. What is the output of `list(map(lambda x: x+1, [1,2,3]))`?

1. `[1,2,3]`
2. `[2,3,4]`
3. `[1,4,9]`
4. Error

Answer 2 · `[2,3,4]`

ADVANCED TOPICS

35. Which keyword is used to handle exceptions?

1. catch
2. except
3. error
4. final

Answer 2 · `except`

ADVANCED TOPICS

36. What happens if an exception is not caught?

1. Python ignores it
2. The program may terminate with a traceback
3. It is converted to a warning
4. The function retries automatically

Answer 2 · The program may terminate with a traceback

ADVANCED TOPICS

37. Which block always runs whether an exception occurs or not?

1. try
2. except
3. finally
4. raise

Answer 3 · finally

ADVANCED TOPICS

38. What does raise do in Python?

1. Suppresses an error
2. Creates a loop
3. Triggers an exception
4. Imports an exception module

Answer 3 · Triggers an exception

ADVANCED TOPICS

39. Which statement about modules is correct?

1. A module can contain functions and variables
2. A module must contain a class
3. Modules cannot be imported twice
4. Modules only store constants

Answer 1 · A module can contain functions and variables

ADVANCED TOPICS

40. What is the purpose of virtual environments in Python?

1. To speed up loops
2. To isolate project dependencies
3. To replace functions with lambdas
4. To compile Python to C

Answer 2 · To isolate project dependencies

ADVANCED TOPICS

41. Which built-in function is often used with iterables to apply a function to each item?

1. filter()
2. map()
3. reduce()
4. zip()

Answer 2 · map()

ADVANCED TOPICS

42. What does filter(function, iterable) return?

1. A sorted list
2. An iterator of items for which the function is true
3. A dictionary
4. A string

Answer 2 · An iterator of items for which the function is true

ADVANCED TOPICS

43. Which function combines elements from multiple iterables pairwise?

1. map()
2. filter()
3. zip()
4. enumerate()

Answer 3 · zip()

ADVANCED TOPICS

44. What is the output of list(zip([1,2], ['a','b']))?

1. [(1,'a'), (2,'b')]
2. [[1,'a'], [2,'b']]
3. {'1':'a', '2':'b'}
4. (1,'a',2,'b')

Answer 1 · [(1,'a'), (2,'b')]

ADVANCED TOPICS

45. What does enumerate(['x','y']) produce?

1. Only the values
2. Pairs of index and value
3. Only the indexes
4. A sorted list

Answer 2 · Pairs of index and value

ADVANCED TOPICS

46. What is the output of list(enumerate(['a','b'], start=1))?

1. [(0,'a'), (1,'b')]
2. [(1,'a'), (2,'b')]
3. [(1,'b'), (2,'a')]
4. [('a',1), ('b',2)]

Answer 2 · [(1,'a'), (2,'b')]

ADVANCED TOPICS

47. Which keyword is used to create a generator function?

1. yield
2. return
3. generate
4. iter

Answer 1 · yield

ADVANCED TOPICS

48. How is a generator different from a list?

1. A generator stores all values immediately
2. A generator produces values lazily
3. A list cannot be iterated
4. A generator cannot be used in loops

Answer 2 · A generator produces values lazily

ADVANCED TOPICS

49. Which statement about Python decorators is most accurate?

1. They permanently change Python syntax
2. They modify or extend function behavior
3. They are only used in Pandas
4. They replace exceptions

Answer 2 · They modify or extend function behavior

ADVANCED TOPICS

50. What is the likely result of deep copying versus shallow copying a nested list?

1. Both always behave identically
2. Deep copy duplicates inner objects, shallow copy may share them
3. Shallow copy duplicates everything recursively
4. Deep copy only works on tuples

Answer 2 · Deep copy duplicates inner objects, shallow copy may share them

DATA TYPES AND OPERATORS

51. What is the output of `print(3 ** 2 ** 1)` in Python?

1. 9
2. 6
3. 8
4. 12

Answer 1 · 9

DATA TYPES AND OPERATORS

52. Which expression evaluates to True?

1. `5 == '5'`
2. `5 != 5.0`
3. `5 <= 5`
4. `'abc' > 'abd'`

Answer 3 · `5 <= 5`

DATA TYPES AND OPERATORS

53. What is the result of `10 / 3` in Python 3?

1. 3
2. 3.0
3. 3.33
4. 3.3333333333333335

Answer 4 · 3.3333333333333335

54. Which operator is used for floor division?

1. /
2. //
3. %
4. **

Answer 2 · //

55. What is the value of True + True * False?

1. 0
2. 1
3. 2
4. True

Answer 2 · 1

56. Which of the following is an immutable Python data type?

1. List
2. Dictionary
3. Set
4. Tuple

Answer 4 · Tuple

57. What is the output of print(type(4.0))?

1. <class 'int'>
2. <class 'float'>
3. <class 'double'>
4. float

Answer 2 · <class 'float'>

58. What does the % operator return?

1. Quotient
2. Exponent
3. Remainder
4. Decimal part

Answer 3 · Remainder

59. Which expression results in False?

1. not False
2. True and 1
3. 0 or True
4. bool("")

Answer 4 · bool("")

DATA TYPES AND OPERATORS

60. What is the output of print(7 % 4)?

1. 1
2. 2
3. 3
4. 4

Answer 3 · 3

DATA TYPES AND OPERATORS

61. Which statement about strings is correct?

1. Strings are mutable
2. Strings can only store letters
3. Strings are immutable
4. Strings cannot be indexed

Answer 3 · Strings are immutable

DATA TYPES AND OPERATORS

62. What is the result of 'Udacity'[:3]?

1. Uda
2. dacity
3. aci
4. Udac

Answer 1 · Uda

DATA TYPES AND OPERATORS

63. What is the output of print('5' * 2)?

1. 10
2. 25
3. 55
4. Error

Answer 3 · 55

DATA TYPES AND OPERATORS

64. Which built-in function can be used to find the length of a string?

1. count()
2. size()
3. len()
4. length()

Answer 3 · len()

DATA TYPES AND OPERATORS

65. What is the output of print(4 == 4.0)?

1. False
2. True
3. Error
4. 4.0

Answer 2 · True

DATA STRUCTURES

66. Which data structure stores key-value pairs?

1. List
2. Tuple
3. Dictionary
4. Set

Answer 3 · Dictionary

DATA STRUCTURES

67. Which of the following allows duplicate values?

1. Set
2. Dictionary keys
3. List
4. Frozen set

Answer 3 · List

DATA STRUCTURES

68. What is the output of `len({1, 2, 2, 3})`?

1. 4
2. 3
3. 2
4. Error

Answer 2 · 3

DATA STRUCTURES

69. Which method adds an element to the end of a list?

1. `insert()`
2. `append()`
3. `extendone()`
4. `add()`

Answer 2 · `append()`

DATA STRUCTURES

70. What is the result of `[1, 2] + [3, 4]`?

1. `[1, 2, 3, 4]`
2. `[4, 6]`
3. Error
4. `[[1,2],[3,4]]`

Answer 1 · `[1, 2, 3, 4]`

DATA STRUCTURES

71. Which data structure is unordered and contains unique elements?

1. List
2. Tuple
3. Set
4. Dictionary

Answer 3 · Set

DATA STRUCTURES

72. What is the output of {'a': 1, 'b': 2}['a']?

1. a
2. 1
3. b
4. Error

Answer 2 · 1

DATA STRUCTURES

73. Which method can be used to remove a key from a dictionary and return its value?

1. delete()
2. remove()
3. pop()
4. discard()

Answer 3 · pop()

DATA STRUCTURES

74. What happens if you try to use a list as a dictionary key?

1. It works normally
2. It is converted to tuple
3. Raises TypeError
4. Returns None

Answer 3 · Raises TypeError

DATA STRUCTURES

75. Which slicing expression returns the last element of a list a?

1. a[0]
2. a[-1]
3. a[1]
4. a[:-1]

Answer 2 · a[-1]

DATA STRUCTURES

76. What is the output of sorted([3, 1, 2])?

1. [3, 2, 1]
2. [1, 2, 3]
3. (1, 2, 3)
4. Error

Answer 2 · [1, 2, 3]

DATA STRUCTURES

77. Which method is used to add multiple items from one list to another?

1. append()
2. extend()
3. insert()
4. join()

Answer 2 · extend()

DATA STRUCTURES

78. What is the output of list('abc')?

1. ['abc']
2. ['a', 'b', 'c']
3. ('a', 'b', 'c')
4. Error

Answer 2 · ['a', 'b', 'c']

DATA STRUCTURES

79. Which of the following is true about tuples?

1. They are mutable
2. They are unordered
3. They are immutable
4. They store only numbers

Answer 3 · They are immutable

DATA STRUCTURES

80. What is the output of (1, 2, 3)[1:]?

1. (1)
2. (2, 3)
3. [2, 3]
4. Error

Answer 2 · (2, 3)

CONTROL FLOW

81. Which keyword is used to start a conditional branch when previous conditions fail?

1. for
2. elif
3. def
4. try

Answer 2 · elif

CONTROL FLOW

82. What is the output of if 0: print('A') else: print('B')?

1. A
2. B
3. 0
4. Error

Answer 2 · B

CONTROL FLOW

83. How many times will for i in range(2, 8, 2) iterate?

1. 2
2. 3
3. 4
4. 5

Answer 2 · 3

CONTROL FLOW

84. Which loop is best when the number of iterations is unknown beforehand?

1. for
2. while
3. if
4. elif

Answer 2 · while

CONTROL FLOW

85. What is the output of `sum(range(1, 5))`?

1. 10
2. 15
3. 6
4. 5

Answer 1 · 10

CONTROL FLOW

86. What does `break` do inside a loop?

1. Skips current iteration
2. Ends the loop immediately
3. Restarts the loop
4. Pauses execution

Answer 2 · Ends the loop immediately

CONTROL FLOW

87. What does `continue` do inside a loop?

1. Ends the loop
2. Skips to the next iteration
3. Repeats the same iteration
4. Exits the program

Answer 2 · Skips to the next iteration

CONTROL FLOW

88. What is the first value produced by `range(3)`?

1. 1
2. 0
3. 3
4. -1

Answer 2 · 0

CONTROL FLOW

89. What is the output of `for i in range(3): print(i, end=' ')`?

1. 1 2 3
2. 0 1 2
3. 0 1 2 3
4. 3 2 1

Answer 2 · 0 1 2

CONTROL FLOW

90. Which statement is true about Boolean conditions in Python?

1. Only True and False can be tested
2. Empty strings evaluate to True
3. Zero can evaluate to False
4. Non-empty lists evaluate to False

Answer 3 · Zero can evaluate to False

FUNCTIONS

91. Which keyword is used to define a function in Python?

1. func
2. define
3. def
4. lambda

Answer 3 · def

FUNCTIONS

92. What is returned by a function with no explicit return statement?

1. 0
2. False
3. None
4. Empty string

Answer 3 · None

FUNCTIONS

93. What is the output of `def f(x): return x * 2` followed by `print(f(3))`?

1. 5
2. 6
3. 9
4. Error

Answer 2 · 6

FUNCTIONS

94. What is a parameter in a function definition?

1. The returned value
2. A variable in the function header
3. A function call
4. A built-in function

Answer 2 · A variable in the function header

FUNCTIONS

95. What is the difference between a parameter and an argument?

1. No difference
2. Parameter is passed in call and argument is in definition
3. Parameter is in definition and argument is passed in call
4. Arguments are only numbers

Answer 3 · Parameter is in definition and argument is passed in call

FUNCTIONS

96. Which built-in function converts a value to an integer when possible?

1. `str()`
2. `float()`
3. `int()`
4. `bool()`

Answer 3 · `int()`

FUNCTIONS

97. What is the output of `def add(a, b=2): return a+b` then `add(3)`?

1. 2
2. 3
3. 5
4. Error

Answer 3 · 5

FUNCTIONS

98. Which statement about scope is correct?

1. Variables inside a function are always global
2. Variables defined inside a function are local by default
3. Global variables cannot be read inside functions
4. Local variables can be accessed everywhere

Answer 2 · Variables defined inside a function are local by default

FUNCTIONS

99. What is the output of `def f(x=[]): x.append(1); return x` after calling `f()` twice?

1. `[1]` then `[1]`
2. `[1]` then `[1, 1]`
3. `[]` then `[1]`
4. Error

Answer 2 · `[1]` then `[1, 1]`

FUNCTIONS

100. Which function can display help information about Python objects in the interpreter?

1. `info()`
2. `doc()`
3. `help()`
4. `manual()`

Answer 3 · `help()`